R7020E · OBJECT DETECTION AND LOCALIZATION · GROUP 7

Fire extinguisher detection, decoy rejection and 3D localization

RGB-D camera · YOLO detector · depth-based real/decoy filter · 3D position in metres

TASK 1 · DETECTION

Kelechi Athanesious Ezekwe

YOLO11n detector, trained on Roboflow photos

TASK 2 · REAL VS DECOY

Shravani Kuchi

Depth-based filter that rejects printed decoys

TASK 3 · 3D + EVALUATION

Thanigaiarasu Shanmugam

3D centre, tracking, evaluation pipeline

October 2026

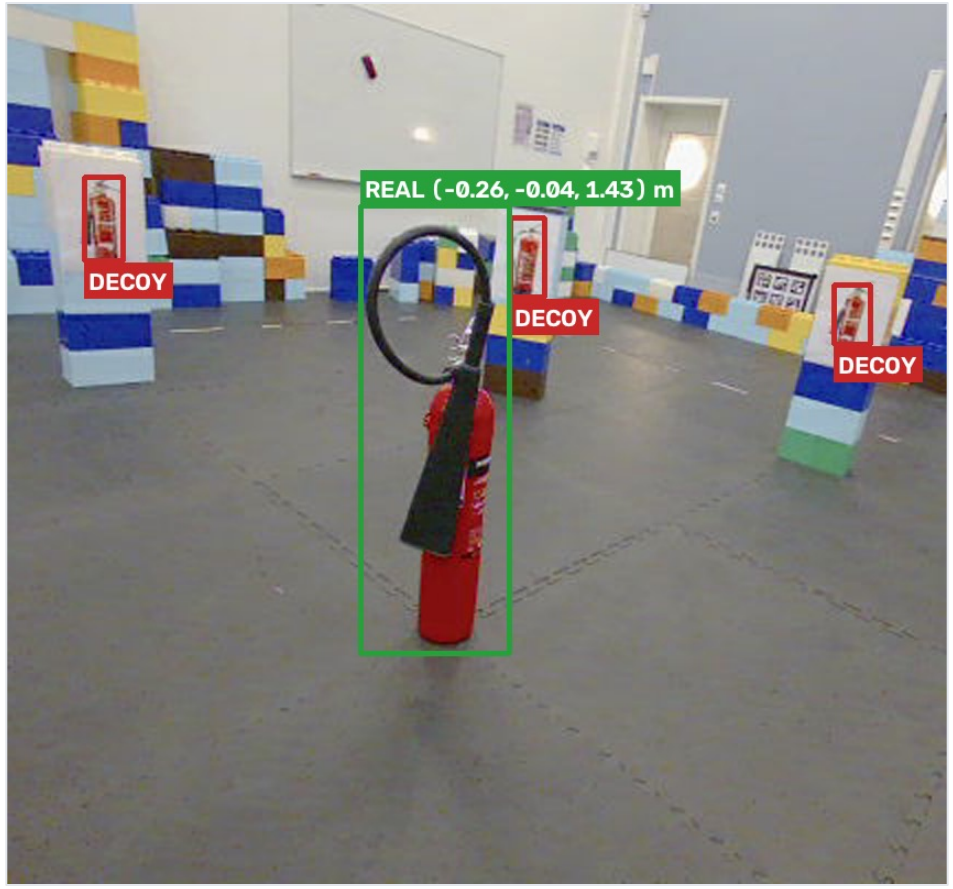
MOTIVATION

Real or printed? A robot must know the difference

Real-world problem: robots must tell real fire extinguishers from printed decoys.

Core challenge: in a colour image a printed extinguisher looks just like a real one.

Our idea: **use depth (RGB-D)** — a print is flat on a wall or block, a real one stands free in 3D.



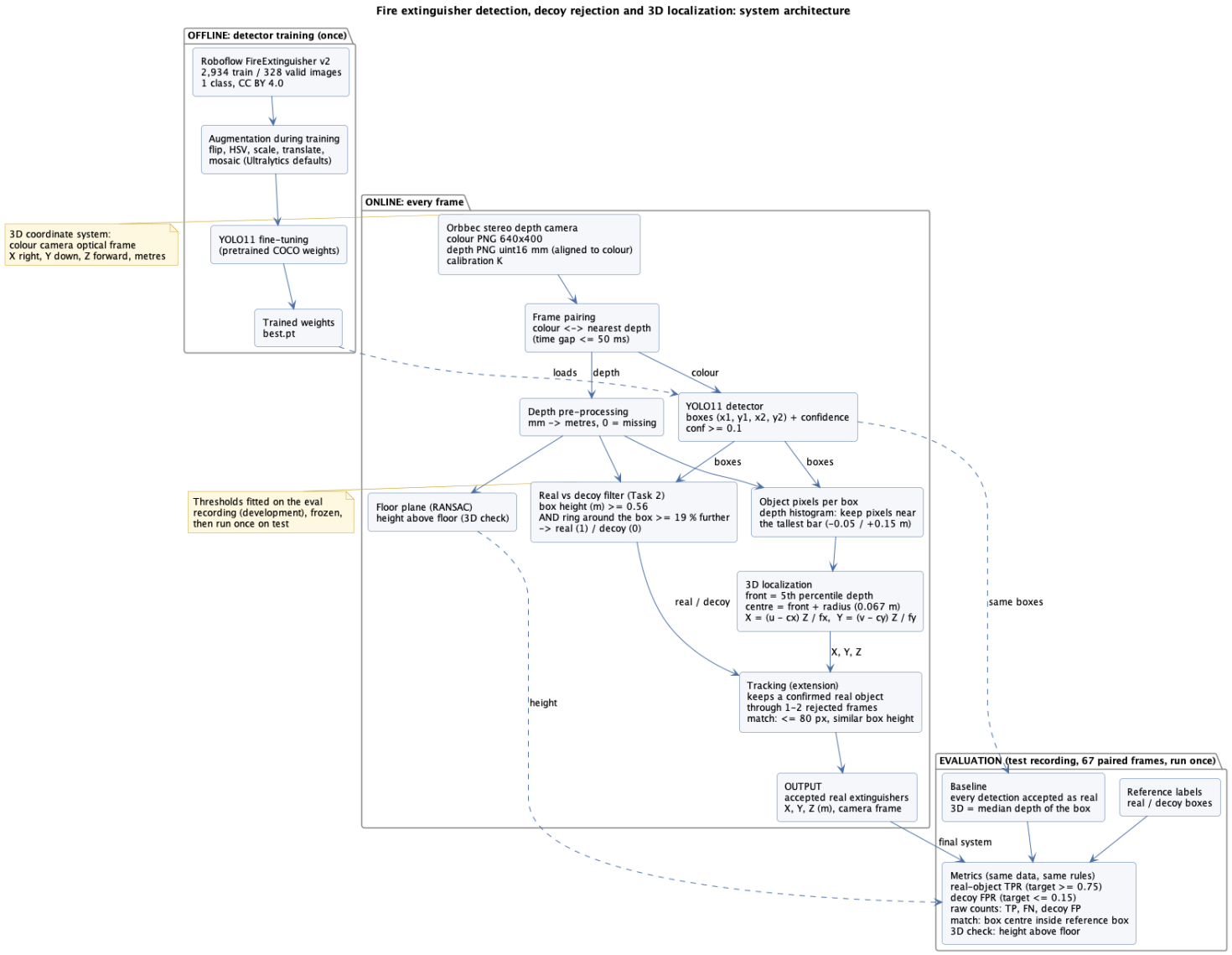
Same scene: one real extinguisher, several printed decoys

Why it matters

- **Safety** compliance inspection
- **Emergency** response
- **Indoor robot** navigation

SYSTEM ARCHITECTURE

One pipeline, three stages



1 Detection

Kelechi

YOLO11n finds every candidate box (conf ≥ 0.10)

2 Real vs decoy

Shravani

box height ≥ 0.56 m AND background ≥ 19 % further

3 3D + tracking

Thanigaiarasu

X, Y, Z of the object centre in metres

whole pipeline **≈ 66 FPS**

Kelechi Athanesious Ezekwe

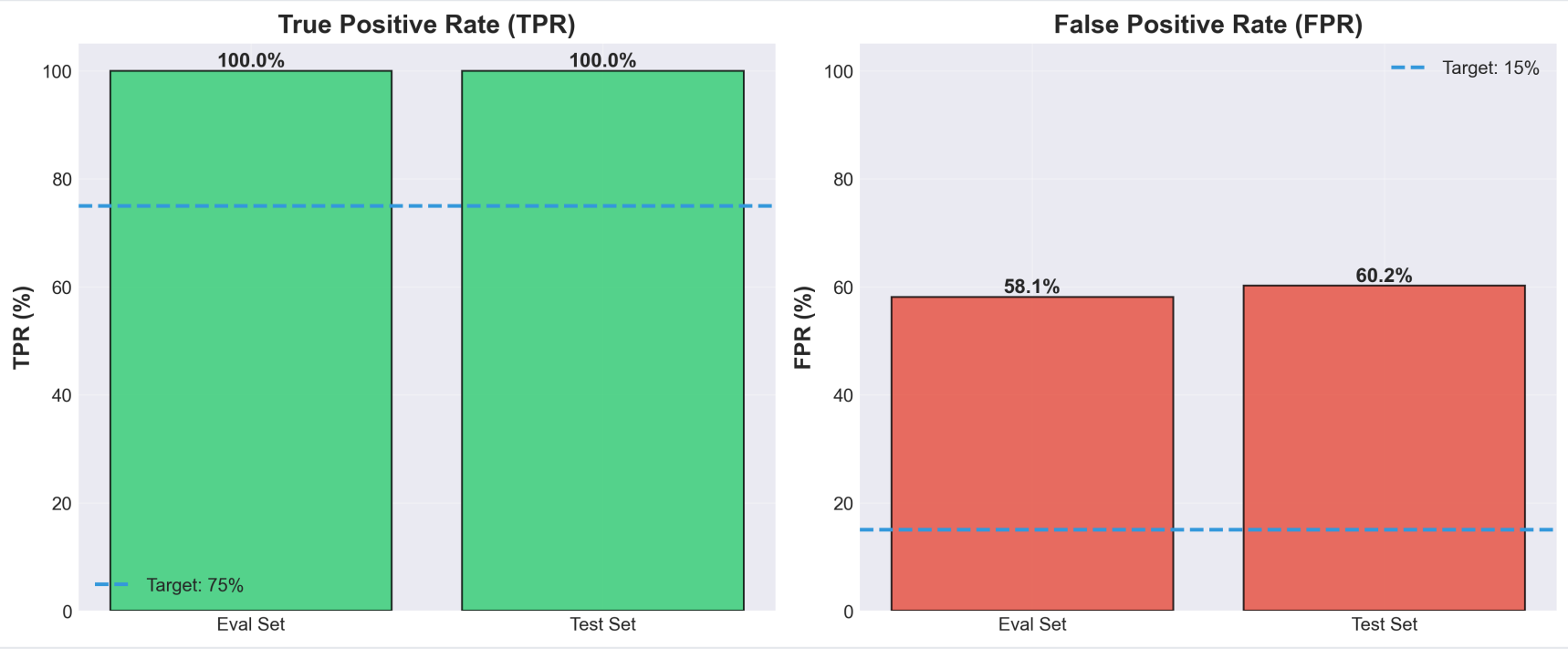
YOLO11n detector, trained on Roboflow photos only

1 Roboflow photos
2,934 train / 328 val
1 class, web photos

2 Fine-tune YOLO11n
pretrained on COCO
100 epochs, patience 20

3 Default augmentation
flip, HSV, scale,
translate, mosaic

4 Low threshold
keep boxes with
confidence ≥ 0.10



0.93
mAP50 on Roboflow validation

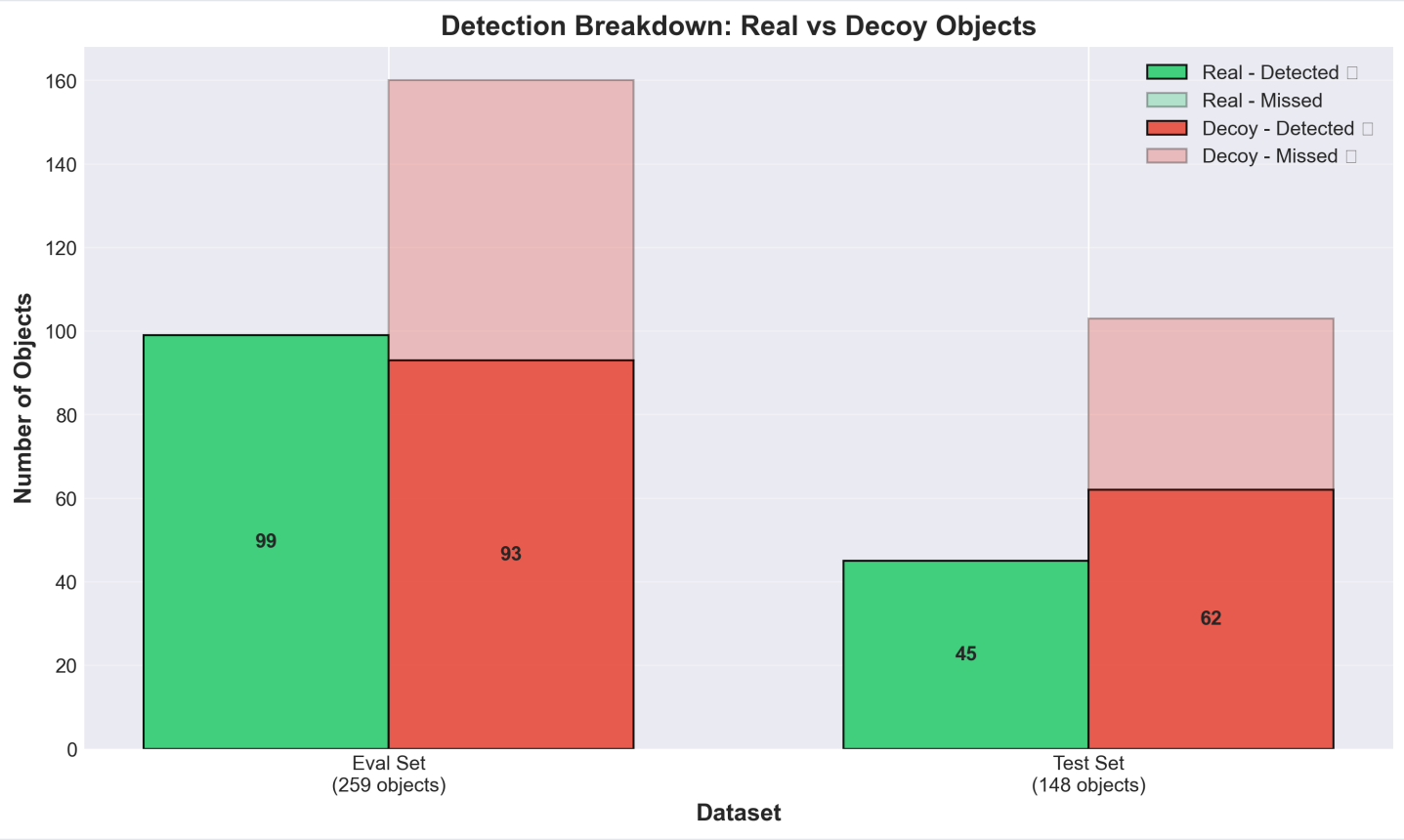
1.00
TPR: every real found (eval and test)

0.58 / 0.60
decoys also detected (eval / test)

14.6 ms
per frame for detection

Kelechi Athanesious Ezekwe

Detector alone: every real found, but most decoys too



	eval	test
Real detected	99 / 99	45 / 45
Decoys detected	93 / 160	62 / 103
Decoy rate	0.58	0.60

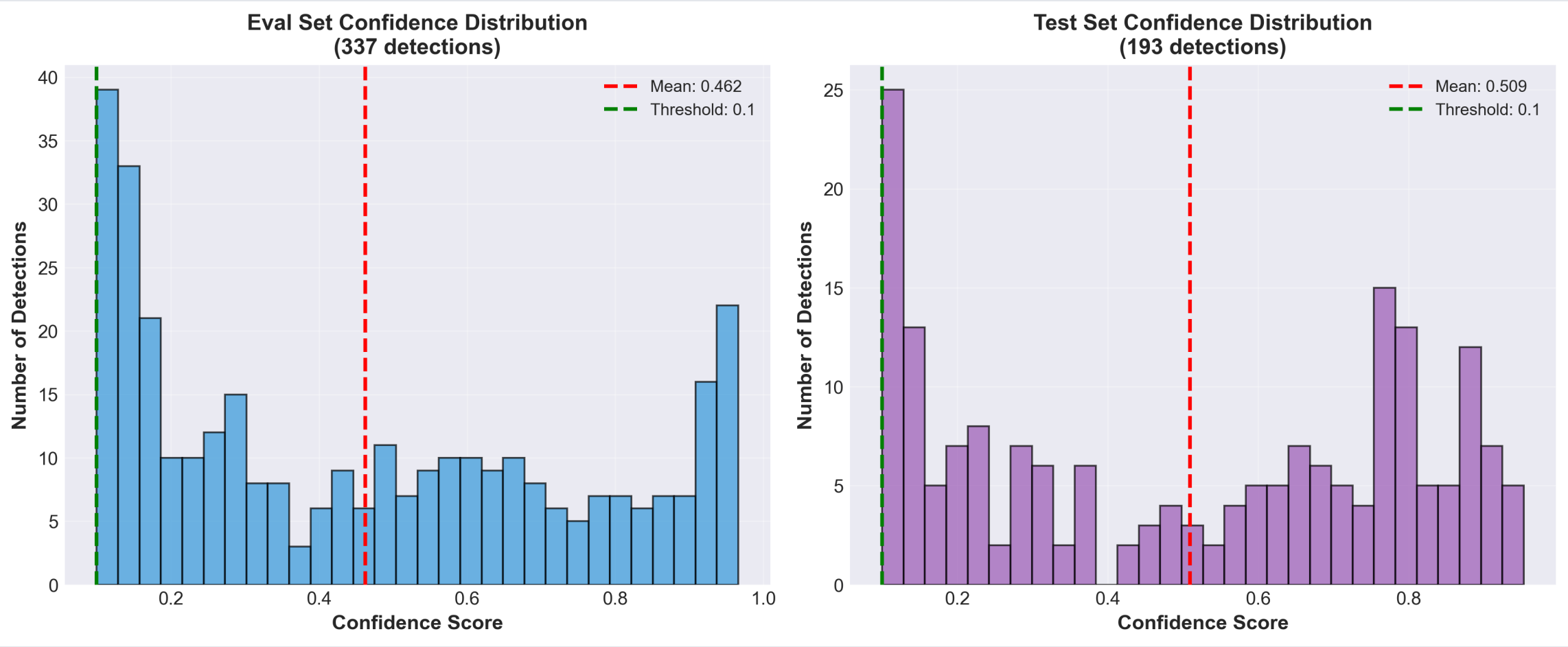
Printed decoys **look real in colour**

→ **Stage 2** must remove them

This detector-only result is the **baseline**

Kelechi Athanesious Ezekwe

Low threshold (0.10) on purpose



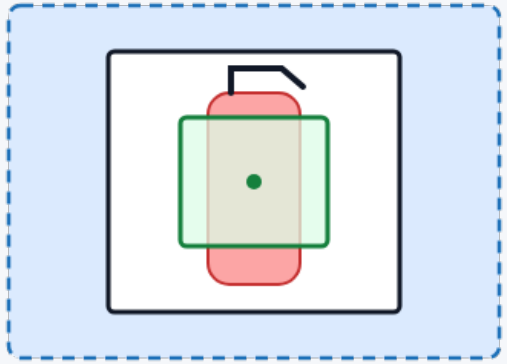
All detector boxes at **conf ≥ 0.10** (real, decoy, background)

A real missed here is **lost for good**

Decoys are removed later by **Stage 2**

Shravani Kuchi

Two depth cues from the detector box



- **Outer ring (+50 % box pad):** median valid depth
- **Detector box:** depth frame within 50 ms
- **Central half:** 25th percentile of valid depth

1 Projected box height (m)

$$H_{\text{proj}} = \frac{h_{\text{box}} \cdot Z_{\text{core},25\%}}{f_y}$$

Metric height represented by the detection box

2 Relative depth contrast

$$\Delta Z_{\text{rel}} = \frac{Z_{\text{ring,med}} - Z_{\text{core},25\%}}{Z_{\text{core},25\%}}$$

Compares the ring depth with the central estimate

SAVED RULE (FINAL FIT ON EVAL DATA)

height ≥ 0.5554 m AND contrast ≥ 0.1922

REAL if both pass

Both cues must be valid

A printed decoy is **small** in metres and sits **flat** on its background

Shravani Kuchi

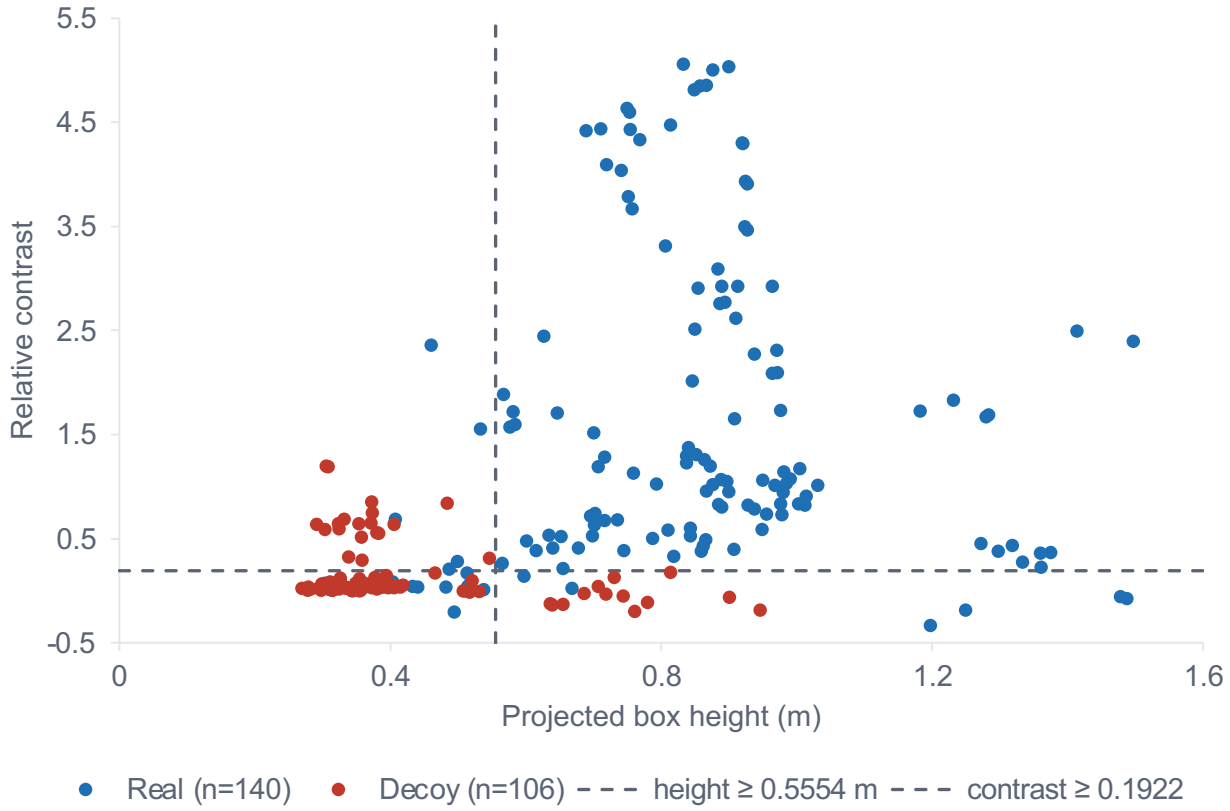
Threshold selection on eval, checked on held-out blocks

01 Reference match
box centre inside a reference box

02 Three time blocks
fit on two, check the held-out one

03 Threshold search
real TPR ≥ 0.75, then fewest decoys

04 Final eval fit
fit on all eval, save for test



Held-out check (eval)

Cutoffs fitted on two blocks, applied to the third

Block	Real accepted	Decoys accepted
1	31 / 33	0 / 53
2	33 / 33	1 / 55
3	33 / 33	1 / 52
Total	**97 / 99**	**2 / 160**

Accept-all baseline

93 / 160

decoys · TPR 1.00 · FPR 0.58

Held-out filter

2 / 160

decoys · TPR 0.98 · FPR 0.013

Why did two decoys pass? Each held-out block uses cutoffs fitted on the other two. The final fit on all eval data rejects both.

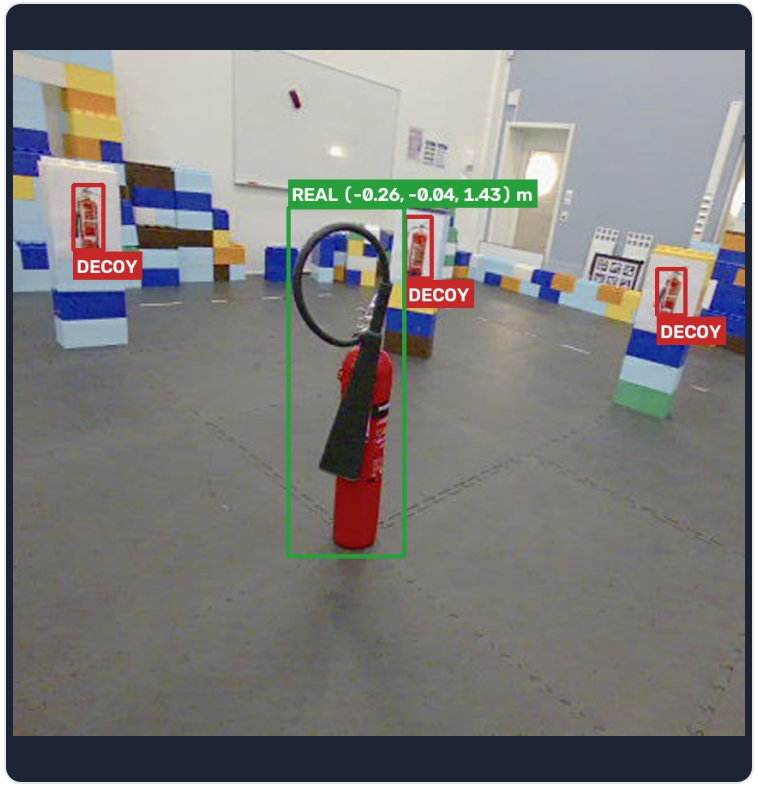
Shravani Kuchi

Test examples and boundary cases



✓ **Accepted real object**

both cues pass



✓ **Rejected printed decoys**

small and flat against the blocks



✗ **Real object rejected**

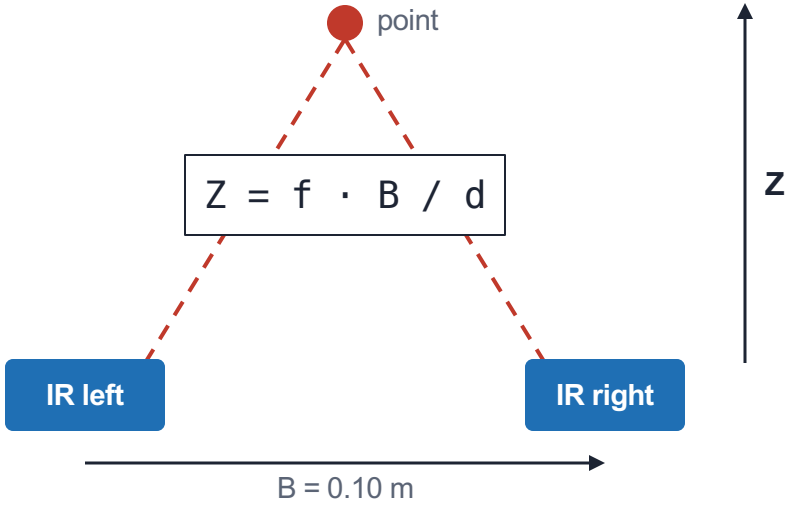
height 0.955 m, contrast 0.098 < 0.192

Test: **2 real missed**, **0 / 103 decoys** accepted

Mixed or missing depth near a box can change **either cue**

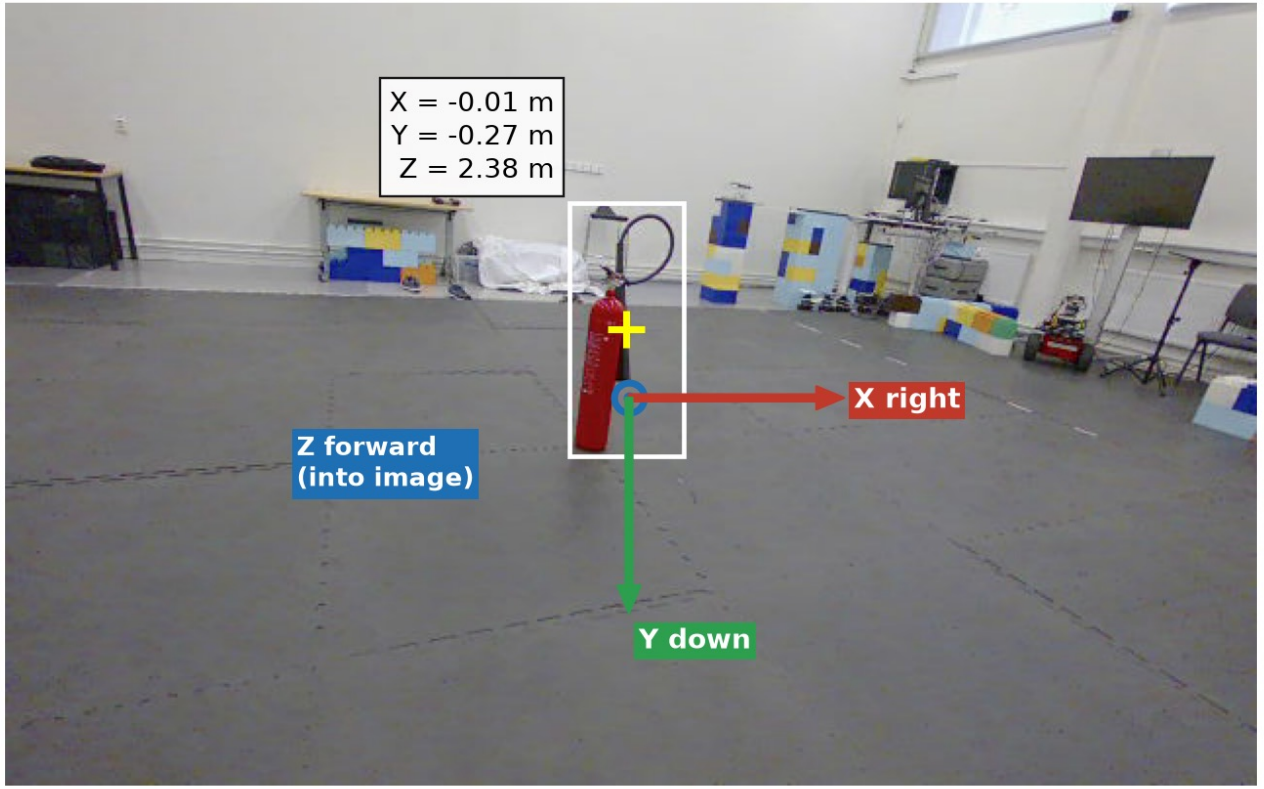
Thanigaiarasu Shanmugam

Depth from stereo, position in the camera frame



- dense disparity **d** per pixel (Lab 4)
- depth **aligned to colour**
- colour ↔ depth pairs **≤ 50 ms**
- **f ≈ 306 px** (camera_info)
- frame: **X right, Y down, Z forward**, metres

Colour camera optical frame, origin at the camera, metres



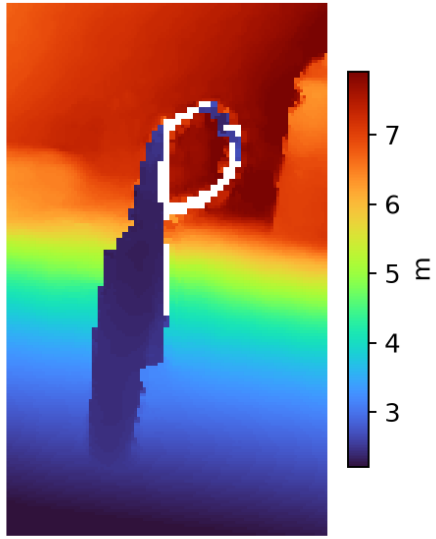
Thanigaiarasu Shanmugam

From box to 3D centre

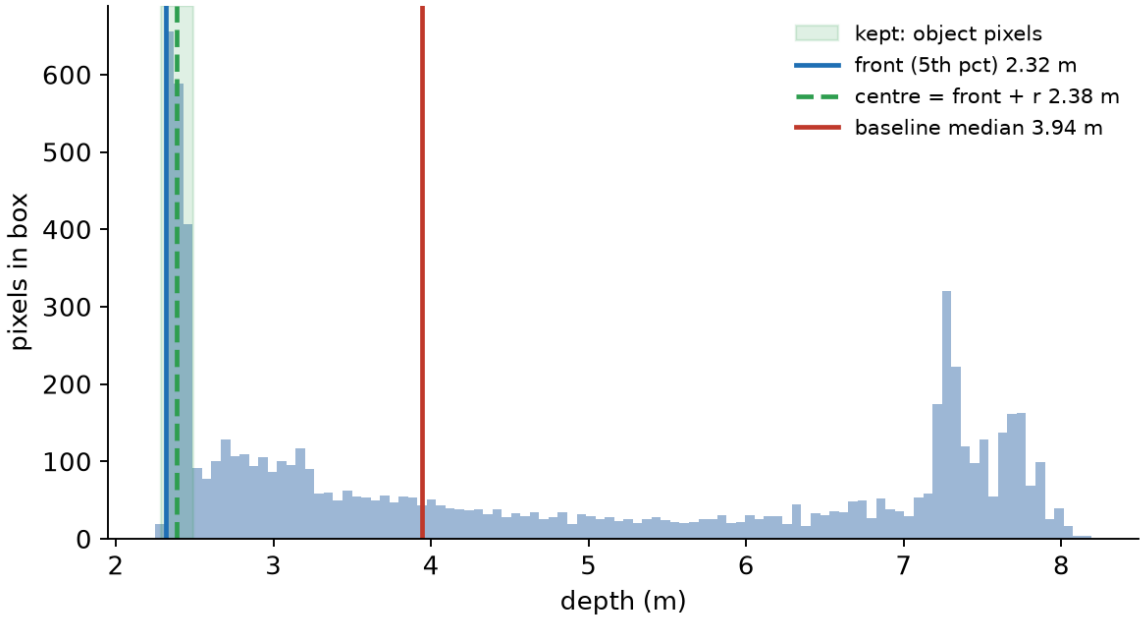
1. YOLO box (colour)



2. depth in the box



3. tallest bar = object; wall/floor thrown away



1 Tallest histogram bar = object
keep −5 / +15 cm

2 Front = 5th percentile
robust to noisy pixels

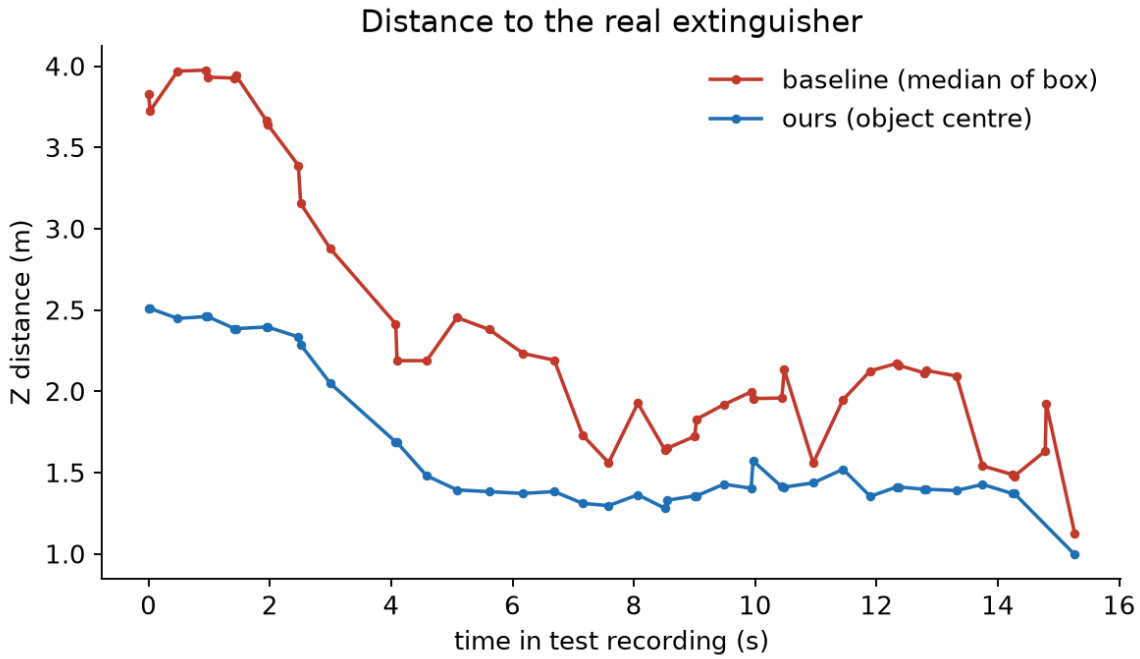
3 Centre = front + r
r = 6.7 cm (cylinder)

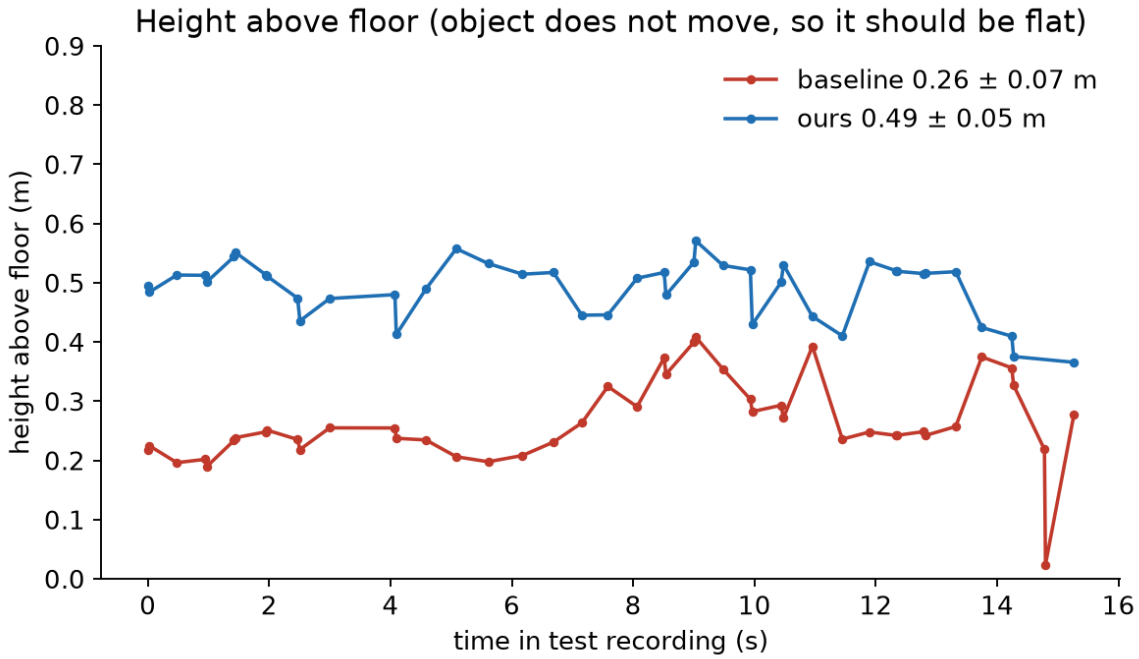
4 Back-project box centre
`X=(u−cx)Z/f · Y=(v−cy)Z/f`

tracking extension: keeps a confirmed real through **≤ 2 rejected frames** (≤ 80 px, chosen on eval)

Thanigaiarasu Shanmugam

3D estimates over the test recording (final evaluation)





0.49 ± 0.05 m

ours: height above floor (≈ half object height)

0.26 ± 0.07 m

baseline: median includes the wall

t (s)	ours X, Y, Z (m)	baseline Z (m)
0.0	−0.08, −0.29, 2.51	3.83
8.1	−0.18, 0.03, 1.36	1.93
11.4	−0.17, 0.07, 1.52	1.95

No 3D ground truth: a stable height above the floor is the consistency check

EVALUATION · 1 / 3

Evaluation setup

Roboflow
2,934 train / 328 val photos
train YOLO

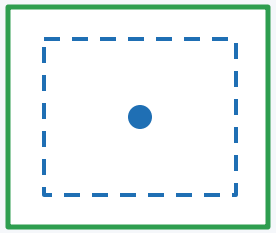
→

eval recording
99 frames · 99 real / 160 decoy
development: all choices

→

test recording
67 frames · 45 real / 103 decoy
evaluation: locked, run once

Match rule



reference box

detection
centre inside
→ **match**

Metrics (per object, per frame)

TPR = real accepted / all real (target ≥ 0.75)

FPR = decoys accepted / all decoys (target ≤ 0.15)

Thresholds

detector conf **≥ 0.10**

real if box height **≥ 0.56 m**

AND background **≥ 19 %** further than object

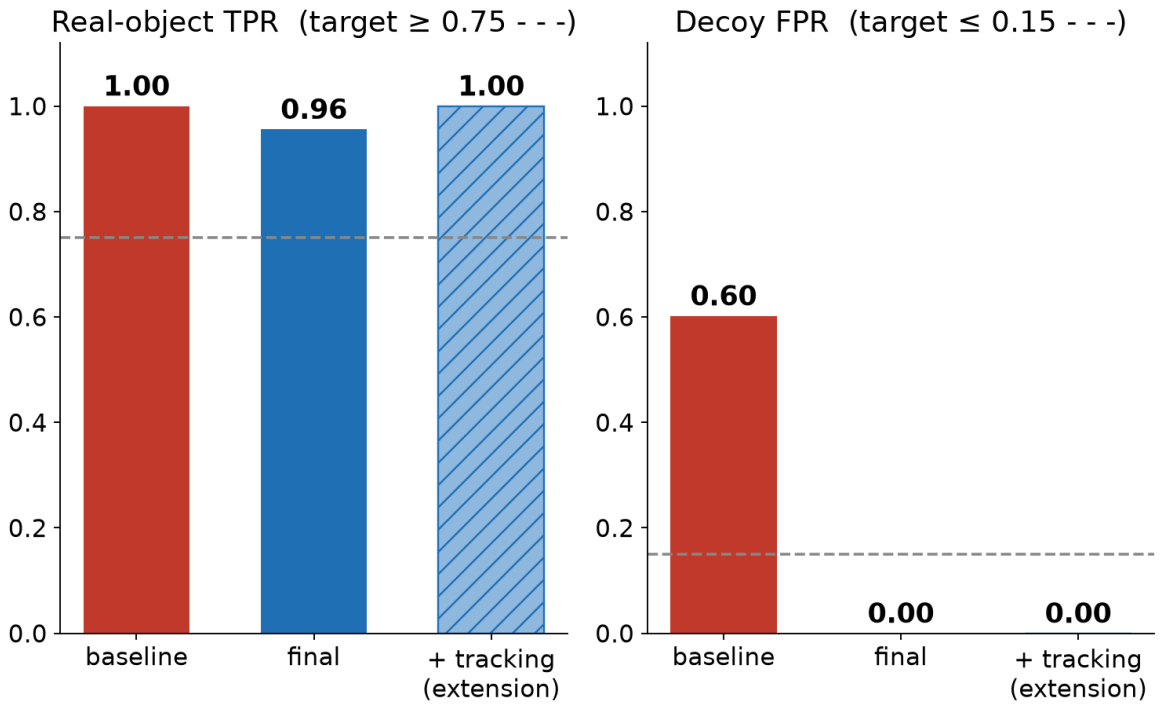
fitted on eval (3 time blocks, cross-validated)

Reference labels: YOLO proposals reviewed frame by frame (**AI-assisted**) + team spot-check

Same data and same rules for **baseline** and **final**

EVALUATION · 2 / 3

Baseline vs final (test, same data)



	baseline	final	+ tracking*
Real: accepted / missed	45 / 0	**43 / 2**	45 / 0
Decoys accepted	62 / 103	**0 / 103**	0 / 103
TPR	1.00	**0.96**	1.00
Decoy FPR	0.60	**0.00**	0.00
Background accepted	22	**2**	2
Height above floor (m)	0.26 ± 0.07	**0.49 ± 0.05**	0.48 ± 0.06
Speed	—	**≈ 66 FPS**	≈ 66 FPS

Baseline = same YOLO, **accept all**, median depth

Trade-off: **−2 real** for **−62 decoys**

Both targets met

* extension: settings chosen on eval (80 px, 2 frames)

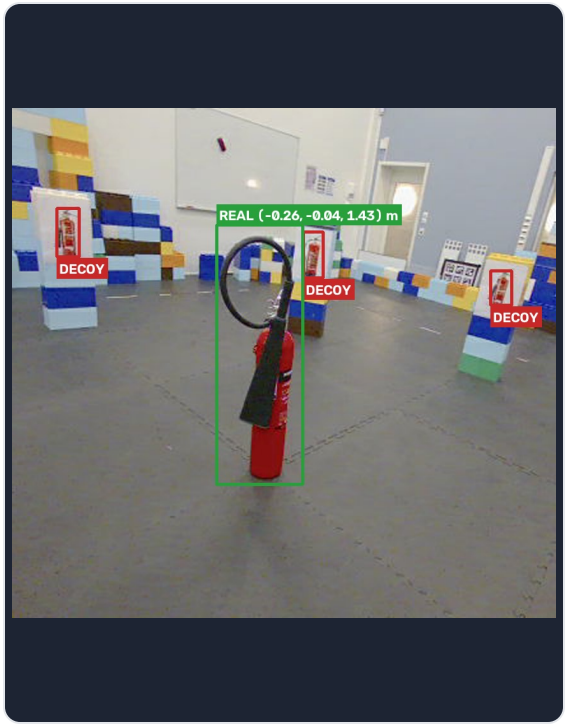
EVALUATION · 3 / 3

Visual evidence (test)



✓ **Real accepted + 3D**

with its position in metres



✓ **Decoys rejected**

prints on blocks: no free space around



✗ **Real missed by filter**

camera turning: depth 28 ms older → box sees floor



✓ **Same frame + tracking***

seen as real in earlier frames → kept

Decoy accepted as real: **none** (0 / 103)

Filter misses: 2 (same turning moment, one old depth frame) → both kept by tracking*

Tracking adds **no** decoys or background boxes

SUMMARY

Final result and limitations

0.96
TPR, final system (target ≥ 0.75)

0.00
decoy FPR: 0 of 103 accepted (target ≤ 0.15)

1.00
TPR with the tracking extension

≈ 66 FPS
15.2 ms per frame, all stages

What worked

- Simple depth cues remove every printed decoy
- 3D centre from the object pixels, not the box median
- Every choice made on eval; test run once at the end
- Tracking recovers the 2 misses without adding decoys

Limitations

- Small test set: 67 frames, one real extinguisher
- Fast camera turns: colour/depth time gap → 2 misses
- No 3D ground truth: checked by stable height only
- Thresholds fitted in one room; new scenes may need a refit

Questions?